Unix　学习报告

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打开终端，在确保当前文件夹内没有文件夹 test2 的前提下新建文件夹 test2，用于本次实验，并使用 cd 命令切换入 test2 文件夹，结果如下图：

```
tpx@tpx-virtual-machine:~$ ls
Desktop    Downloads  hello.c  Music     Public  Templates  test.txt
Documents  hello      mpi      Pictures  snap    test       Videos
tpx@tpx-virtual-machine:~$ mkdir test2
tpx@tpx-virtual-machine:~$ ls
Desktop    Downloads  hello.c  Music     Public  Templates  test2     Videos
Documents  hello      mpi      Pictures  snap    test       test.txt
tpx@tpx-virtual-machine:~$ cd test2
```

1. 熟悉 diff 命令：

   使用 touch 创建 a.txt,b.txt，并使用 vim 编辑两文件。

```
tpx@tpx-virtual-machine:~/test2$ touch a.txt
tpx@tpx-virtual-machine:~/test2$ ls
a.txt
tpx@tpx-virtual-machine:~/test2$ touch b.txt
tpx@tpx-virtual-machine:~/test2$ ls
a.txt  b.txt
```

a.txt 的内容如下图：

```
tpx@tpx-virtual-machine: ~/test2
int main()
{
char a='c';
int b=10;
printf("%d",b);
retunr 0;
}
~
~
~
~
~
~
~
```

b.txt 的内容如下图：

```
tpx@tpx-virtual-machine: ~/test2
main() int
{
char a='c';
int b=10;
printf("%d",b);
}
~
~
```

使用命令 diff a.txt b.txt 的结果如下:

```
tpx@tpx-virtual-machine:~/test2$ diff a.txt b.txt
1c1
<  int main()
---
> main() int
6d5
< return 0;
```

使用命令 diff -c a.txt b.txt 的结果如下:

```
tpx@tpx-virtual-machine:~/test2$ diff -c a.txt b.txt
*** a.txt       2025-03-31 14:19:15.266794783 +0800
--- b.txt       2025-03-31 14:18:47.097776316 +0800
***************
*** 1,7 ****
!  int main()
  {
  char a='c';
  int b=10;
  printf("%d",b);
- return 0;
  }
--- 1,6 ----
! main() int
  {
  char a='c';
  int b=10;
  printf("%d",b);
  }
```

使用命令 diff -u a.txt b.txt 的结果如下:

```
tpx@tpx-virtual-machine:~/test2$ diff -u a.txt b.txt
--- a.txt       2025-03-31 14:19:15.266794783 +0800
+++ b.txt       2025-03-31 14:18:47.097776316 +0800
@@ -1,7 +1,6 @@
- int main()
+main() int
 {
 char a='c';
 int b=10;
 printf("%d",b);
-return 0;
 }
```

以并排形式输出：

```
tpx@tpx-virtual-machine:~/test2$ diff -y -W 40 a.txt b.txt
 int main()          |    main() int
{                         {
char a='c';               char a='c';
int b=10;                 int b=10;
printf("%d",b);           printf("%d",b);
return 0;            <
}                         }
```

2.

使用 patch 命令进行两个文件的对比和两个文件夹的对比以及补丁：由所学内容，当欲将 a.txt 修改为 b.txt 时，应使用 diff a.txt b.txt 生成补丁文件，再使用使 patch 命令将补丁文件应用到 a.txt 中。

仍使用上个实验中中的 test2 文件夹以及 a.txt b.txt 文件，并使用重定向将 diff 命令的补丁文件输入到 patch 命令中

首先，查看修改前 a.txt,b.txt 中的内容，将 diff 命令的结果重定向到 change.txt 中，再将 change.txt 作为 patch 命令的输入：

```
tpx@tpx-virtual-machine:~/test2$ cat a.txt
 int main()
{
char a='c';
int b=10;
printf("%d",b);
return 0;
}
tpx@tpx-virtual-machine:~/test2$ cat b.txt
main() int
{
char a='c';
int b=10;
printf("%d",b);
}
tpx@tpx-virtual-machine:~/test2$ touch change.txt
tpx@tpx-virtual-machine:~/test2$ diff a.txt b.txt >change.txt
tpx@tpx-virtual-machine:~/test2$ patch a.txt<change.txt
patching file a.txt
tpx@tpx-virtual-machine:~/test2$ cat a.txt
main() int
{
char a='c';
int b=10;
printf("%d",b);
}
```

使用 diff 命令生成两个文件夹 t1，t2 的补丁文件，并使用补丁文件和 patch 命令将 t1 修改为 t2.结果如下图：

```
tpx@tpx-virtual-machine:~/test2$ ls
a.txt  b.txt  change.txt  rtygrep.txt  t1  t2
tpx@tpx-virtual-machine:~/test2$ cd t1
tpx@tpx-virtual-machine:~/test2/t1$ ls
t11  t12
tpx@tpx-virtual-machine:~/test2/t1$ cd t11
tpx@tpx-virtual-machine:~/test2/t1/t11$ touch t111.txt
tpx@tpx-virtual-machine:~/test2/t1/t11$ vim t111.txt
tpx@tpx-virtual-machine:~/test2/t1/t11$ cd ../
tpx@tpx-virtual-machine:~/test2/t1$ cd t12
tpx@tpx-virtual-machine:~/test2/t1/t12$ ls
tpx@tpx-virtual-machine:~/test2/t1/t12$ touch t122.txt
tpx@tpx-virtual-machine:~/test2/t1/t12$ vim t122.txt
tpx@tpx-virtual-machine:~/test2/t1/t12$ cd ../
tpx@tpx-virtual-machine:~/test2/t1$ ls
t11  t12
tpx@tpx-virtual-machine:~/test2/t1$ touch t1s.txt
tpx@tpx-virtual-machine:~/test2/t1$ vim t1s.txt
tpx@tpx-virtual-machine:~/test2/t1$ cd ../
tpx@tpx-virtual-machine:~/test2$ cd t2
tpx@tpx-virtual-machine:~/test2/t2$ ls
t21  t2_1.txt
tpx@tpx-virtual-machine:~/test2/t2$ vim t2_1.txt
tpx@tpx-virtual-machine:~/test2/t2$ cd t21
tpx@tpx-virtual-machine:~/test2/t2/t21$ touch tt21.txt
tpx@tpx-virtual-machine:~/test2/t2/t21$ vim tt21.txt
tpx@tpx-virtual-machine:~/test2/t2/t21$ ls
tt21.txt
tpx@tpx-virtual-machine:~/test2/t2/t21$ cd ../
tpx@tpx-virtual-machine:~/test2/t2$ ls
t21  t2_1.txt
tpx@tpx-virtual-machine:~/test2/t2$ cd ../
tpx@tpx-virtual-machine:~/test2$ ls
a.txt  b.txt  change.txt  rtygrep.txt  t1  t2
tpx@tpx-virtual-machine:~/test2$ diff -Nur t1 t2 > t1t2.patch
```

```
tpx@tpx-virtual-machine:~/test2$ cat t1t2.patch
diff -Nur t1/t11/t111.txt t2/t11/t111.txt
--- t1/t11/t111.txt     2025-03-31 19:49:23.471213561 +0800
+++ t2/t11/t111.txt     1970-01-01 08:00:00.000000000 +0800
@@ -1 +0,0 @@
-this is t111.txt
diff -Nur t1/t12/t122.txt t2/t12/t122.txt
--- t1/t12/t122.txt     2025-03-31 19:50:21.556522530 +0800
+++ t2/t12/t122.txt     1970-01-01 08:00:00.000000000 +0800
@@ -1 +0,0 @@
-ithis is t122.txt
diff -Nur t1/t1s.txt t2/t1s.txt
--- t1/t1s.txt  2025-03-31 19:51:23.288239115 +0800
+++ t2/t1s.txt  1970-01-01 08:00:00.000000000 +0800
@@ -1 +0,0 @@
-this is t1s.txt
diff -Nur t1/t21/tt21.txt t2/t21/tt21.txt
--- t1/t21/tt21.txt     1970-01-01 08:00:00.000000000 +0800
+++ t2/t21/tt21.txt     2025-03-31 19:52:36.035768709 +0800
@@ -0,0 +1,2 @@
+this is tt21.txt
+
diff -Nur t1/t2_1.txt t2/t2_1.txt
--- t1/t2_1.txt 1970-01-01 08:00:00.000000000 +0800
+++ t2/t2_1.txt 2025-03-31 19:51:59.378619941 +0800
@@ -0,0 +1 @@
+this is t2_1.txt
tpx@tpx-virtual-machine:~/test2$ cd t1
tpx@tpx-virtual-machine:~/test2/t1$ ls
t11  t12  t1s.txt
tpx@tpx-virtual-machine:~/test2/t1$ cd t11
tpx@tpx-virtual-machine:~/test2/t1/t11$ ls
t111.txt
tpx@tpx-virtual-machine:~/test2/t1/t11$ cd ../
tpx@tpx-virtual-machine:~/test2/t1$ cd  t12
tpx@tpx-virtual-machine:~/test2/t1/t12$ ls
t122.txt
tpx@tpx-virtual-machine:~/test2/t1/t12$ cd ../
```

```
tpx@tpx-virtual-machine:~/test2/t1$  patch -p1<../t1t2.patch
patching file t11/t111.txt
patching file t12/t122.txt
patching file t1s.txt
patching file t21/tt21.txt
patching file t2_1.txt
tpx@tpx-virtual-machine:~/test2/t1$ ls
t21  t2_1.txt
tpx@tpx-virtual-machine:~/test2/t1$ cd t21
tpx@tpx-virtual-machine:~/test2/t1/t21$ ls
tt21.txt
```

3. 使用 cut 命令获取用户名和 shell 路径，结果如图：

```
tpx@tpx-virtual-machine:~/test2$ cut -d: -f 1,7 /etc/passwd
root:/bin/bash
daemon:/usr/sbin/nologin
bin:/usr/sbin/nologin
sys:/usr/sbin/nologin
sync:/bin/sync
games:/usr/sbin/nologin
man:/usr/sbin/nologin
lp:/usr/sbin/nologin
mail:/usr/sbin/nologin
news:/usr/sbin/nologin
uucp:/usr/sbin/nologin
proxy:/usr/sbin/nologin
www-data:/usr/sbin/nologin
backup:/usr/sbin/nologin
list:/usr/sbin/nologin
irc:/usr/sbin/nologin
gnats:/usr/sbin/nologin
nobody:/usr/sbin/nologin
systemd-network:/usr/sbin/nologin
systemd-resolve:/usr/sbin/nologin
messagebus:/usr/sbin/nologin
systemd-timesync:/usr/sbin/nologin
syslog:/usr/sbin/nologin
_apt:/usr/sbin/nologin
tss:/bin/false
uuidd:/usr/sbin/nologin
systemd-oom:/usr/sbin/nologin
tcpdump:/usr/sbin/nologin
avahi-autoipd:/usr/sbin/nologin
usbmux:/usr/sbin/nologin
```

```
systemd-resolve:/usr/sbin/nologin
messagebus:/usr/sbin/nologin
systemd-timesync:/usr/sbin/nologin
syslog:/usr/sbin/nologin
_apt:/usr/sbin/nologin
tss:/bin/false
uuidd:/usr/sbin/nologin
systemd-oom:/usr/sbin/nologin
tcpdump:/usr/sbin/nologin
avahi-autoipd:/usr/sbin/nologin
usbmux:/usr/sbin/nologin
dnsmasq:/usr/sbin/nologin
kernoops:/usr/sbin/nologin
avahi:/usr/sbin/nologin
cups-pk-helper:/usr/sbin/nologin
rtkit:/usr/sbin/nologin
whoopsie:/bin/false
sssd:/usr/sbin/nologin
speech-dispatcher:/bin/false
fwupd-refresh:/usr/sbin/nologin
nm-openvpn:/usr/sbin/nologin
saned:/usr/sbin/nologin
colord:/usr/sbin/nologin
geoclue:/usr/sbin/nologin
pulse:/usr/sbin/nologin
gnome-initial-setup:/bin/false
hplip:/bin/false
gdm:/bin/false
tpx:/bin/bash
sshd:/usr/sbin/nologin
another:/bin/bash
dennis:/bin/bash
daniel:/bin/bash
abigale:/bin/bash
```

4. 使用 grep 命令进行匹配，首先生成 rtygrep.txt 文件,作为 grep 命令的参数如下：

```
tpx@tpx-virtual-machine:~/test2$ touch rtygrep.txt
tpx@tpx-virtual-machine:~/test2$ vim rtygrep.txt
tpx@tpx-virtual-machine:~/test2$ cat rtygrep
cat: rtygrep: No such file or directory
tpx@tpx-virtual-machine:~/test2$ cat rtygrep.txt
test
goop
doop
look.
hook1573
tast
3de1998
76hu1998
213e.
juhtv
hellooooaaa
hellooaas
gooog
ahuug00gqwe
```

```
tpx@tpx-virtual-machine:~/test2$ grep -E '\<t[ea]st\>' rtygrep.txt
test
tast
tpx@tpx-virtual-machine:~/test2$ grep -v 'goo' rtygrep.txt
test
doop
look.
hook1573
tast
3de1998
76hu1998
213e.
juhtv
hellooooaaa
hellooaas
ahuug00gqwe
```

```
tpx@tpx-virtual-machine:~/test2$ grep '[0-9]' rtygrep.txt
hook1573
3de1998
76hu1998
213e.
ahuug00gqwe
tpx@tpx-virtual-machine:~/test2$ grep '^[^a-zA-Z]' rtygrep.txt
3de1998
76hu1998
213e.
tpx@tpx-virtual-machine:~/test2$ grep '\.$' rtygrep.txt
look.
213e.
tpx@tpx-virtual-machine:~/test2$ grep -E '^[35]..1998$' rtygrep.txt
3de1998
tpx@tpx-virtual-machine:~/test2$ grep -v '^48' rtygrep.txt
test
goop
doop
look.
hook1573
tast
3de1998
76hu1998
213e.
juhtv
hellooaaa
hellooaas
gooog
ahuug00gqwe
```

```
tpx@tpx-virtual-machine:~/test2$ grep -E 'o{2}' rtygrep.txt
goop
doop
look.
hook1573
hellooaaa
hellooaas
gooog
tpx@tpx-virtual-machine:~/test2$ grep -E 'go{2,5}g' rtygrep.txt
gooog
```

P332 应用题

P332.1：date 输出重定向到 currentdate 文件

```
tpx@tpx-virtual-machine:~/test2$ date>currentdate.txt
tpx@tpx-virtual-machine:~/test2$ cat currentdate.txt
2025年 03月 31日 星期一 20:15:25 CST
```

P332.2:

```
tpx@tpx-virtual-machine:~/test2$ users|tee userlist.txt|wc -w
1
tpx@tpx-virtual-machine:~/test2$ cat userlist.txt
tpx
```

P332.3:

```
tpx@tpx-virtual-machine:~/test2$ sort /etc/passwd|tee userids.txt|wc -l
53
tpx@tpx-virtual-machine:~/test2$ cat userids.txt
abigale:x:1004:1002:abigale,,,:/home/abigale:/bin/bash
another:x:1001:1001:another,oo,om,om,om:/home/another:/bin/bash
_apt:x:105:65534::/nonexistent:/usr/sbin/nologin
avahi-autoipd:x:110:119:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/s
nologin
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
colord:x:123:130:colord colour management daemon,,,:/var/lib/colord:/usr/sbi
login
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-he
:/usr/sbin/nologin
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
daniel:x:1003:1002:daniel,,,:/home/daniel:/bin/bash
dennis:x:1002:1002:dennis,n,,:/home/dennis:/bin/bash
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
fwupd-refresh:x:120:126:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
games:x:5:60:games:/usr/games:/usr/sbin/nologin
gdm:x:128:134:Gnome Display Manager:/var/lib/gdm3:/bin/false
geoclue:x:124:131::/var/lib/geoclue:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/no
n
gnome-initial-setup:x:126:65534::/run/gnome-initial-setup/:/bin/false
hplip:x:127:7:HPLIP system user,,,:/run/hplip:/bin/false
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,,:/:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
messagebus:x:102:105::/nonexistent:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
```

P332.4:

应附加 2>/dev/null 将错误信息输出到空设备文件中（标准错误输出重定向到空设备文件中） 不会显示到屏幕上

P352 2 应用题:

p352.2.1: 直接在 wc -l 后跟多个文本文件名，或先用 cat 命令合并 5 个文本文件，在用管道连接 wc -l

```
tpx@tpx-virtual-machine:~/test2$ wc -l data1.txt data2.txt data3.txt data4.txt data5.txt
  9 data1.txt
 16 data2.txt
 24 data3.txt
 33 data4.txt
 22 data5.txt
104 total
tpx@tpx-virtual-machine:~/test2$ cat data1.txt data2.txt data3.txt data4.txt data5.txt|wc -l
104
tpx@tpx-virtual-machine:~/test2$ ls
data1.txt  data2.txt  data3.txt  data4.txt  data5.txt  rtygrep.txt
```

P352.2.2:

```
tpx@tpx-virtual-machine:~/test2$ cat important.txt
hello.
i am
well that doesn't matter.
have a nice day!
tpx@tpx-virtual-machine:~/test2$ tac important.txt
have a nice day!
well that doesn't matter.
i am
hello.
tpx@tpx-virtual-machine:~/test2$ rev important.txt
.olleh
 ma i
.rettam t'nseod taht llew
!yad ecin a evah
tpx@tpx-virtual-machine:~/test2$ rev important.txt|tac
!yad ecin a evah
.rettam t'nseod taht llew
 ma i
.olleh
```

P352.2.3:

使用 sort 命令代替 dmesg 命令，并使用 tail -n 25 仅显示其输出的后 25 行（管道连接 sort 和 tail，两次管道为了说明确实这样操作的结果确实只有 25 行）

```
tpx@tpx-virtual-machine:~/test2$ sort /etc/passwd|tail -n 25|wc -l
25
tpx@tpx-virtual-machine:~/test2$ sort /etc/passwd|tail -n 25
nm-openvpn:x:121:127:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
pulse:x:125:132:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
root:x:0:0:root:/root:/bin/bash
rtkit:x:116:123:RealtimeKit,,,:/proc:/usr/sbin/nologin
saned:x:122:129::/var/lib/saned:/usr/sbin/nologin
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
sshd:x:129:65534::/run/sshd:/usr/sbin/nologin
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
syslog:x:104:111::/home/syslog:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-oom:x:108:117:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
tcpdump:x:109:118::/nonexistent:/usr/sbin/nologin
tpx:x:1000:1000:tpx,,,:/home/tpx:/bin/bash
tss:x:106:113:TPM software stack,,,:/var/lib/tpm:/bin/false
usbmux:x:111:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
uuidd:x:107:116::/run/uuidd:/usr/sbin/nologin
whoopsie:x:117:124::/nonexistent:/bin/false
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
```